(guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

ANNUAL REVIEWS

Annual Review of Animal Biosciences

Translational Relevance and Future Integration of the Oncopig Cancer Model in Preclinical Applications

Lobna Elkhadragy,[1] Ron C. Gaba,[1,2] Matthew M. Niemeyer,[1] Lawrence B. Schook,[1,3,4] and Kyle M. Schachtschneider[3]

[1]Department of Radiology, University of Illinois at Chicago, Chicago, Illinois, USA; email: lobna@uic.edu, rgaba@uic.edu, mniemeye@uic.edu

[2]Department of Pathology, University of Illinois at Chicago, Chicago, Illinois, USA

[3]Sus Clinicals, Inc., Chicago, Illinois, USA; email: kyle@susclinicals.com

[4]Department of Animal Sciences, University of Illinois at Urbana-Champaign, Urbana, Illinois, USA; email: schook@illinois.edu

ANNUAL REVIEWS CONNECT

www.annualreviews.org

- Download figures
- Navigate cited references
- Keyword search
- Explore related articles
- Share via email or social media

Annu. Rev. Anim. Biosci. 2025. 13:465–81

First published as a Review in Advance on October 17, 2024

The *Annual Review of Animal Biosciences* is online at animal.annualreviews.org

https://doi.org/10.1146/annurev-animal-111523-102125

Copyright © 2025 by the author(s). This work is licensed under a Creative Commons Attribution 4.0 International License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited. See credit lines of images or other third-party material in this article for license information.

OPEN ACCESS

Keywords

Oncopig Cancer Model, Oncopig, transgenic pig model, pig cancer model, large animal model

Abstract

Porcine cancer models offer a valuable platform for evaluating interventions such as devices, surgeries, and locoregional therapies, which are often challenging to test in mouse models. In addition to size and anatomical similarities with humans, pigs share greater similarities in genetics, immunity, drug metabolism, and metabolic rate with humans as compared to mouse models, increasing their translational relevance. This review focuses on the Oncopig Cancer Model, a genetically engineered porcine model designed to recapitulate human cancer. Harboring a transgenic cassette that expresses oncogenic mutant *KRAS* and *TP53* under control of a Cre-Lox system, the Oncopig allows temporal and spatial control of tumor induction. Its versatility has enabled the development of diverse cancer models including liver, pancreatic, lung, and bladder cancer. Serving as a clinically relevant model for human cancer, the Oncopig addresses unmet clinical needs and holds immense promise for advancing preclinical cancer research and therapeutic development.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

1. INTRODUCTION

Despite advances in understanding the molecular and immunological landscapes of cancer, coupled with remarkable progress in diagnostics and therapeutics, global cancer incidence and mortality continue to increase rapidly (1). Cancer is a leading cause of death worldwide, accounting for nearly 10 million deaths in 2020 (1). More than 19 million new cancer cases were diagnosed in 2020, and global cancer burden is expected to increase to 28.4 million cases in 2040 (1). Addressing this escalating burden necessitates the development of innovative diagnostic modalities and therapeutic interventions. Preclinical animal models play an indispensable role in this critical pursuit. Bridging the gap between fundamental diagnostic and therapeutic discoveries and human clinical trials, preclinical animal models offer invaluable insights into oncogenesis, tumor biology, screening, detection, immunogenicity, therapeutics, and clinical outcomes, thus driving advancements across the entire cancer research continuum.

Progress in cancer research has been hampered by the lack of suitable animal models that faithfully recapitulate human cancer pathophysiology (2). Ideal models should closely resemble humans in terms of genotype, anatomy, physiology, immunity, metabolism, and epidemiological factors and should develop tumors with molecular features, histological characteristics, and tumor microenvironment similar to those of the human disease. Moreover, it is imperative to simulate cancer development alongside comorbidities commonly observed in cancer patients. These concurrent comorbidities significantly influence patient management, therapeutic approaches, and clinical outcomes, making their incorporation essential for translational relevance in cancer research.

Mouse models are among the most frequently used preclinical cancer models, offering convenient handling, cost-effectiveness, and a wealth of genetic information (3–5). Despite their contributions to cancer research, the translation of findings from mouse studies to successful clinical trials remains a challenge. Only a small percentage of drug candidates identified as potential anticancer treatments in mouse models have proved safe and effective in human clinical trials (2, 4, 5). Differences in genome, metabolism, and immunity between mice and humans contribute to this disparity, underscoring the need for improved predictive models bridging the gap between preclinical research and clinical outcomes (2–5).

Historically, pigs have played a vital role in biomedical research, serving as invaluable models for various human diseases, including cardiovascular, metabolic, and neurodegenerative disorders (6–8). With greater similarities to humans in anatomy, physiology, size, lifespan, immune system, and metabolism compared to rodents, pigs offer immense potential for biomedical, pharmacological, and toxicological investigations (7, 8). Unraveling the porcine genome has provided the foundation for the development of novel animal models to study human diseases and expedite the discovery and validation of drugs, devices, and diagnostics (9, 10). The emergence of animal cloning, genome sequencing, and gene editing has further enhanced pigs' utility as indispensable preclinical translational research models. By harnessing these technological advancements, use of pigs is poised to contribute significantly to the advancement of cancer research and the development of effective therapeutic interventions.

2. THE ONCOPIG CANCER MODEL

The Oncopig Cancer Model is a genetically engineered pig model developed to serve as a large animal platform for cancer research (11). Its conception stemmed from the necessity to address unmet clinical needs through the development of a versatile, genetically defined porcine cancer model. The rationale for its development is that due to pigs' genetic, metabolic, and anatomical

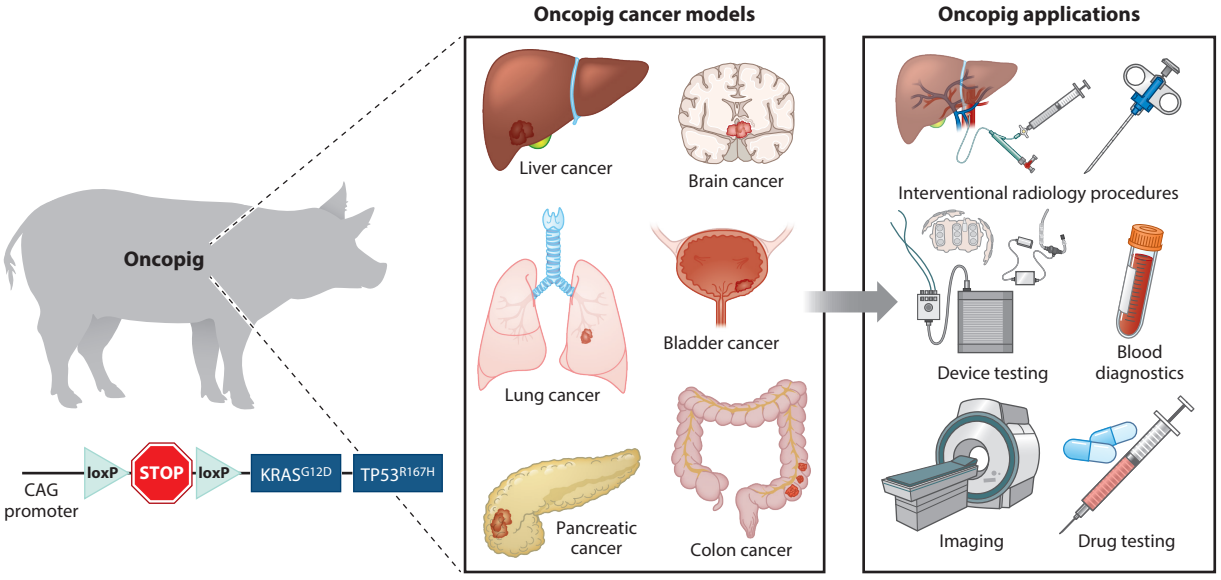

Figure 1

Cancer models developed in the Oncopig and their potential applications. The Oncopig is a transgenic pig with Cre recombinase–inducible expression of $KRAS^{G12D}$ and $TP53^{R167H}$. The transgene construct includes a CAG promoter and Lox-Stop-Lox sequences followed by porcine $KRAS^{G12D}$ and $TP53^{R167H}$. The exposure to Cre recombinase results in site-specific recombination between the two Lox sites, leading to the removal of the stop sequence and induction of transgene expression. Cancer models developed in the Oncopig include liver, lung, pancreatic, brain, bladder, and colon cancers. Given the resemblance between pigs and humans in size, anatomy, physiology, genetics, immunity, and metabolism, the Oncopig offers a valuable platform for testing interventional radiology image–guided procedures such as ablations and arterial delivery and developing imaging modalities and blood diagnostics, device testing, and drug testing.

resemblances to humans, genetic alteration of specific pathways commonly disrupted in human cancers would provide a porcine tumor model reflective of human cancers. The Oncopig, a transgenic pig that harbors genetic alterations analogous to those frequently observed in human cancers, was therefore developed.

The Oncopig harbors a somatic transgenic cassette that expresses oncogenic mutant *KRAS* and *TP53* under control of a Cre-Lox system (**Figure 1**). This construct includes a CAG promoter followed by a Lox-Stop-Lox sequence succeeded by porcine $KRAS^{G12D}$ and $TP53^{R167H}$ transgenes separated by an internal ribosome entry site sequence (11). Both $KRAS^{G12D}$, a constitutively active KRAS mutant, and $TP53^{R167H}$, a dominant-negative mutant TP53, are oncogenic. The exposure to Cre recombinase results in site-specific recombination between the two Lox sites, leading to the removal of the Stop sequence and subsequent induction of transgene expression. As such, the Cre-Lox system enables time- and tissue-specific control of tumor induction in the Oncopig.

By mimicking genetic alterations found in human cancers, the Oncopig provides a valuable platform for investigating various aspects of tumor biology, including initiation, progression, and response to therapies. Compared to traditional rodent models, the Oncopig offers a closer phylogenetic resemblance to humans, enhancing the translational relevance of preclinical cancer research. This review describes the applications of the Oncopig as a cancer model and underscores their transformative potential in advancing translational cancer research.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

3. INDUCTION OF TUMORIGENESIS IN THE ONCOPIG

The validity of the Oncopig for cancer modeling was demonstrated through the induction of tumorigenesis using Cre recombinase, both in vitro and in vivo (11). Fibroblasts isolated from the Oncopig exhibited in vitro hallmarks of tumorigenesis upon treatment with adenovirus encoding Cre recombinase (AdCre) vector (11). These cells expressed the transgenes $KRAS^{G12D}$ and $TP53^{R167H}$ and displayed transformed cell morphology, increased proliferation and migration, and anchorage-independent growth. The tumorigenicity of these cells was further confirmed by the development of tumors in immunocompromised mice, considered the gold standard in vivo assay for cell tumorigenicity.

Consistently, in vivo delivery of AdCre by subcutaneous, intramuscular, or intratesticular injections resulted in tumor development in the Oncopig (11, 12) (**Figure 2**, **Table 1**). Subcutaneous AdCre injection into the flank, ear, abdomen, or neck led to tumor development within 1–3 weeks. Notably, analysis of subcutaneous flank tumors revealed robust infiltration of T cells, with a strong predominance of $CD8\beta^+$ and γδ T cells (12). The infiltrating $CD8\beta^+$ T cells displayed robust expression of perforin and granzyme B, suggesting an antitumor immune response by these cytotoxic immune cells. Additionally, the tumors were enriched with FOXP3-expressing T cells and displayed upregulation of immune checkpoint molecules including indoleamine 2,3-dioxygenase 1, cytotoxic T lymphocyte–associated protein 4, and programmed death ligand 1. These findings indicate the in vivo recognition of the Oncopig tumors by innate and adaptive immune systems, supporting the Oncopig's utility for preclinical testing of immunotherapies. Intramuscular injection of AdCre induced rapid development of mesenchymal tumors resembling leiomyosarcoma, whereas intratesticular injection resulted in poorly differentiated tumors likely originating from sex cord stromal cells (11). Together, these findings demonstrated the validity of the genetically engineered Oncopig Cancer Model as a platform for cancer modeling and paved the way for the development of models for specific cancer types.

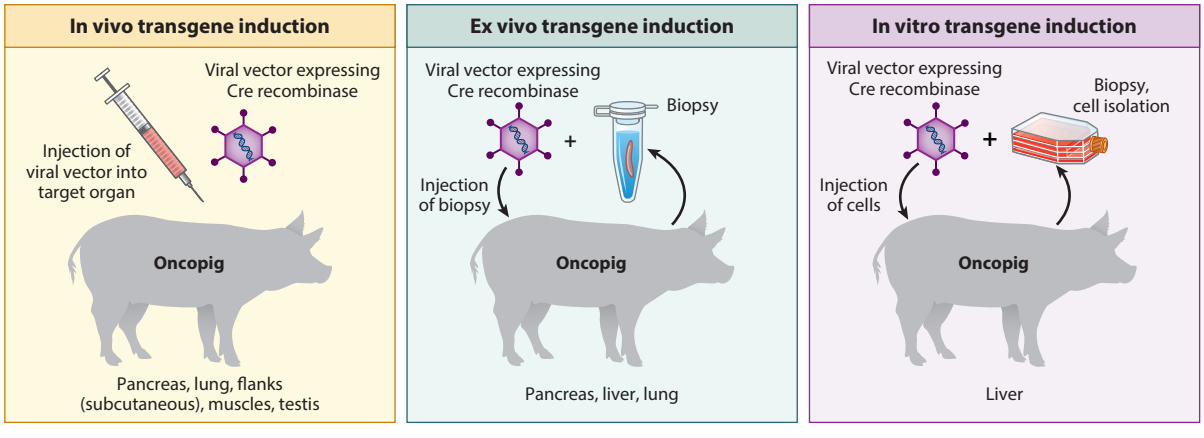

Figure 2

Approaches used to induce transgene expression and tumor development in the Oncopig. The in vivo approach entails direct injection of viral vector expressing Cre recombinase into the target organ. This has been applied for the development of tumors in the pancreas, lung, flanks, muscles, and testis. The ex vivo approach involves obtaining a biopsy from the target organ and incubating it with viral vector that expresses Cre recombinase for a short period of time, followed by reinjection of the dissociated biopsy back into the target organ. This approach has been used to develop pancreatic, liver, and lung tumors. The in vitro approach includes obtaining cells through a biopsy or surgical resection, followed by in vitro exposure to Cre recombinase and culturing of the developed transformed cell line. Next, the cells are reinjected into the target organ in the same Oncopig. This approach has been used to develop liver tumors following liver resection and hepatocyte isolation.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

Table 1 Cancer modeling in Oncopigs

Tumor type/location	Exposure to Cre recombinase	Tumor induction approach	Tumor features	Time for tumor development	References
Subcutaneous tumors	In vivo	Subcutaneous injection of AdCre into flanks, ear, abdomen, or neck tissues.	Tumors with intratumoral T cell infiltration	1–3 weeks	Schook et al. (11), Overgaard et al. (12)
Intramuscular tumors	In vivo	Intramuscular injection of AdCre	Mesenchymal tumor, suggestive of leiomyosarcoma	1–3 weeks	Schook et al. (11)
Intratesticular tumors	In vivo	Intratesticular injection of AdCre	Poorly differentiated tumor	1–3 weeks	Schook et al. (11)
Liver tumors	Ex vivo	Exposure of liver biopsy to AdCre, followed by autologous reinjection into Oncopig liver	Poorly differentiated to undifferentiated carcinomas accompanied by a major inflammatory component	2 weeks	Nurili et al. (14)
	In vitro	Exposure of isolated hepatocytes to AdCre, followed by autologous implantation into Oncopig liver	Poorly differentiated to undifferentiated carcinomas accompanied by a major inflammatory component, recapitulates histological, transcriptional and genomic features of human liver cancer	2 weeks (after implantation)	Nurili et al. (14), Gaba et al. (15)
Liver tumors in subcutaneous tissues	In vitro	Exposure of isolated hepatocytes to AdCre, followed by autologous subcutaneous injection into flanks	Poorly differentiated to undifferentiated carcinomas accompanied by a major inflammatory component	1 week (after implantation)	Elkhadragy et al. (18)
Pancreatic cancer	In vivo	Injection of AdCre into pancreatic duct	Poorly differentiated neoplasms with dense stroma recapitulating human pancreatic cancer, has major inflammatory component	One year	Principe et al. (22)
				3 weeks	Mondal et al. (23)
	Ex vivo	Exposure of pancreatic biopsy to AdCre, followed by autologous reinjection into Oncopig pancreas	Undifferentiated carcinomas, similar to those of the pancreatobiliary system in humans	1 week	Boas et al. (24)
Lung cancer	In vivo, ex vivo	Endovascular injection of AdCre; exposure of lung biopsy to AdCre followed by autologous reinjection into Oncopig lung	Poorly differentiated neoplasm associated with a marked immune response	1 week	Ghosn et al. (25)
Bladder cancer	In vivo	AdCre injection into the bladder	Urothelial cell carcinomas resembling the sarcomatoid variant or inflammatory masses	1 month	Aulitzky et al. (26)

Abbreviations: AdCre, adenovirus expressing Cre recombinase; PDAC, pancreatic ductal adenocarcinoma.

4. CANCER MODELS DEVELOPED IN THE ONCOPIG

The Oncopig has been used to develop models for a wide spectrum of cancers, including liver, pancreatic, lung, bladder, brain, and colorectal cancer (**Table 1**). Approaches to induce tumors in the Oncopig include in vivo, ex vivo, and in vitro exposure to Cre recombinase for induction of transgene expression (**Figure 2**).

4.1. Oncopig Liver Cancer Models

The development of porcine models for liver cancer holds particular significance due to the necessity for large animal models to test locoregional therapies, which are minimally invasive image-guided procedures employed in 50–60% of liver cancer cases (13). Due to their small size and incompatibility with device-based therapeutics, mouse models are not suitable for testing liver-directed locoregional therapies, highlighting the value of large animal liver cancer models. The development of localized tumors measuring 2–3 cm, rather than multinodular diffuse tumors, is preferred for testing locoregional therapies including ablations and arterially delivered therapies.

Liver tumors have been developed in the Oncopig using two different methodologies. The first method is based on ex vivo induction of transgene expression (14), whereas the second method entails in vitro induction of transgene expression in isolated hepatocytes (14, 15). Both methods resulted in the development of localized liver tumors suitable for testing locoregional therapies.

The ex vivo induction method involved obtaining a liver biopsy from the Oncopig under imaging guidance, followed by incubation with AdCre for 20–30 min and subsequent reinjection into the liver parenchyma (14). Due to the large size of the Oncopig liver, up to four sites can be inoculated, facilitating the development of several localized tumors in the same animal. This approach has demonstrated a high success rate (92%) with rapid tumor growth observed within 1–2 weeks, yielding tumors of approximately 2.1 cm in diameter. Tumors were detectable by ultrasound and computed tomography (CT) scan. Histological analysis revealed poorly differentiated to undifferentiated carcinomas accompanied by a major inflammatory component (14).

Conversely, the in vitro induction method involves the isolation of hepatocytes through collagenase digestion of liver tissue obtained by surgical resection or image-guided biopsy. Subsequent in vitro exposure of these hepatocytes to AdCre results in transgene expression and malignant transformation to hepatocellular carcinoma (HCC) (15, 16). Comparative analyses between Oncopig and human HCC cells unveiled similarities in various characteristics, including cell cycle length, migratory capacity, chemotherapeutic susceptibility, and expression of drug-metabolizing enzymes (15). Transcriptional profiling further corroborated these findings, revealing similarity with human HCC transcriptional hallmarks, including TERT reactivation, apoptosis evasion, angiogenesis activation, and Wnt signaling activation (16). Furthermore, xenograft experiments employing subcutaneous injection of Oncopig-derived HCC cells into immunodeficient mice have yielded tumors faithfully recapitulating human HCC, thereby underscoring the fidelity of the Oncopig model in mirroring clinical disease manifestations. Assessment of the tumor microenvironment in these xenografts demonstrated altered expression of genes associated with hepatic fibrosis deposition, immune response, and neoangiogenesis, providing valuable insights into the interplay between HCC cells and their surrounding microenvironment (17). The injection of Oncopig HCC cells into the same pig from which hepatocytes were isolated (autologous approach) resulted in development of subcutaneous and intrahepatic tumors in the Oncopig (14–16). These tumors recapitulated human HCC histological, genomic, and transcriptional features (15, 16). Despite its longer duration and increased labor intensity compared to the ex vivo induction method,

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

the autologous implantation approach offers the advantage of facilitating the introduction of specific gene mutations into cultured cells and enabling the selection of a desirable cell population prior to implantation (15, 18, 19).

4.2. Oncopig Pancreatic Cancer Models

Oncogenic *KRAS* mutations occur in 95% of human pancreatic cancers and coincide with *TP53* mutations in nearly 70% of patients (20, 21), underscoring the relevance of the Oncopig as an ideal large animal model for pancreatic cancer. The anatomical similarity between porcine and human pancreas enables studies, including surgical and locoregional therapies, that are challenging to conduct in murine models. Pancreatic cancer has been induced in the Oncopig both in vitro and in vivo. In vitro exposure of Oncopig pancreatic duct cells to AdCre resulted in their malignant transformation, evidenced by tumor development in immunodeficient mice (22). These transformed cells exhibited ductal lineage markers (E-cadherin and CK19), activated KRAS signaling (phospho-ERK1/2), and increased proliferation. Xenograft tumors developed in mice displayed a dense stromal component reminiscent of human pancreatic ductal adenocarcinoma (PDAC), with cancer cells expressing ductal markers and stromal cells expressing α-smooth muscle actin, a marker of pancreatic stellate cells (22).

The development of in vivo pancreatic cancer in the Oncopig was described in three studies. In two of these studies, AdCre was injected into the main pancreatic duct via a surgical procedure (22, 23). In the study by Principe et al. (22), this approach resulted in the development of small nodules not visible by CT within a year in the Oncopig subjected to the procedure, providing proof-of-principle for pancreatic cancer modeling in the Oncopig. The tumors expressed E-cadherin and CK19 and were accompanied by a dense and desmoplastic tumor stroma rich with leukocytes. Although the predominant histotype recapitulated human PDAC, there were distinct areas resembling human pancreatic neuroendocrine tumors hallmarked by excessive proliferation and desmoplastic stroma. In Mondal et al.'s (23) more recent study, tumors developed in the Oncopigs (10/14, or 71%) within 3 weeks. Several fundamental differences between these two studies can account for the different outcomes, including the latter study using a higher AdCre dose, administration of IL-8 with AdCre to enhance viral entry to epithelial cells, and an elegant surgical modification of the injection technique to increase AdCre retention in the pancreatic duct. The pancreatic tumors developed in this study were poorly differentiated malignant neoplasms displaying increased proliferation, stromal deposition, and immune cell infiltration. Exome and transcriptome analysis of these tumors revealed similarity to the molecular signatures and pathways of human pancreatic cancer. A limitation to this model is the occurrence of gastric outlet obstruction and pancreatitis as a secondary complication caused by rapid pancreatic tumor development and strong inflammatory reaction. Approaches to circumvent these complications will increase the model's utility and allow study of precursor lesions.

Boas et al. (24) used an ex vivo induction approach, wherein a pancreatic biopsy was obtained under imaging guidance and exposed to AdCre ex vivo, followed by autologous injection into the pancreas. This method yielded rapidly developing tumors, with the development of tumors measuring 3 cm in Oncopigs (11/12, or 92%) within one week. These tumors were imaged by CT and histologically resembled undifferentiated carcinomas of the pancreatobiliary tract in humans. Notably, all tumors were accompanied by a major inflammatory component. Although these tumors had a collagen containing desmoplastic stroma, the amount of stroma was significantly less than that of human PDAC (24). Collectively, these studies underscore the utility of the Oncopig as a robust model for studying pancreatic cancer and highlight the need for continued refinement to enhance their translational relevance.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

4.3. Oncopig Lung Cancer Model

The development of large animal models for lung cancer holds immense promise for advancing preclinical investigations of image-guided locoregional therapies and surgical interventions. Recently, a study elucidated the induction of lung neoplasia in the Oncopig through both in vivo and ex vivo transgene induction methodologies (25). In the in vivo approach, AdCre was delivered via endovascular injection through the pulmonary arteries or inferior vena cava in two Oncopigs. Conversely, the ex vivo approach involved the incubation of lung biopsies with AdCre followed by reinjection into the lungs in two Oncopigs. Although both techniques led to the development of lung nodules detectable by CT imaging within a week, the success rates were modest (10% for the in vivo and 33% for the ex vivo approach). Histological analysis of the lung tumors revealed characteristics of undifferentiated neoplasms and a prominent mixed leukocytic infiltrate. Although it demonstrates proof-of-concept for rapid and feasible site-specific induction of lung neoplasia in the Oncopig, this model's main limitation lies in the predominance of undifferentiated tumors, failing to recapitulate the histological diversity observed in human lung cancers, including adenocarcinoma, squamous cell carcinoma, and small cell lung cancer.

4.4. Oncopig Bladder Cancer Model

The Oncopig has emerged as a valuable tool for the development of both in vitro and in vivo models of bladder cancer (26, 27). To establish Oncopig bladder cancer cell lines, urothelial cells were isolated from Oncopig bladders and subjected to AdCre exposure in vitro. These transformed cells exhibited comparable responses to various chemotherapeutic agents, such as cisplatin, doxorubicin, and gemcitabine, akin to human bladder cancer cells (27). Furthermore, an in vivo model of bladder cancer in Oncopigs was developed recently through AdCre injection directly into the bladder. This approach resulted in the development of masses within one month, which were histologically characterized as urothelial cell carcinomas resembling the sarcomatoid variant or inflammatory masses (26). These findings suggest the potential for using Oncopigs for the rapid development of a large animal model of bladder cancer, offering potential avenues for accelerated research in this domain.

5. DEVELOPMENT OF GENETICALLY TAILORED TUMORS USING THE ONCOPIG

With recent advances in gene editing techniques, notably clustered regularly interspaced short palindromic repeats (CRISPR) editing, genetically tailored tumors can be developed in the Oncopig by somatic gene editing. This approach allows the introduction of specific gene mutations in Oncopig tumors, thereby broadening the scope of modeling human tumor mutational profiles. By leveraging somatic gene editing, Oncopigs can serve as a versatile platform for the development of genetically defined tumors, circumventing the need for the laborious and time-consuming process of developing additional genetically engineered pig models harboring distinct gene mutations.

Recent studies have successfully implemented this approach, generating Oncopig subcutaneous and intrahepatic tumors harboring genetically tailored gene mutations (18, 19, 28). In these studies, hepatocytes isolated from Oncopigs were subjected to in vitro transformation with AdCre, followed by CRISPR editing to introduce specific gene mutations. These mutations mimicked frequently occurring alterations observed in human liver cancer, including the knockout of *ARID1A*, *AXIN1*, *PTEN*, and *CDKN2A*. In vitro analyses demonstrated the fidelity of these genetically engineered Oncopig liver cancer cells in recapitulating the effects observed in the human disease. Subsequent autologous injection of CRISPR-edited cells resulted in the development of

subcutaneous and intrahepatic tumors harboring genetically defined gene mutations in Oncopigs (18, 19). These pioneering studies illustrate the feasibility of developing genetically tailored Oncopig tumors, paving the way for transformative research that unravels the contribution of specific gene mutations on tumor progression and treatment response in a large animal model.

6. MODELING CANCER COMORBIDITIES IN THE ONCOPIG

The development of tumors in a clinically relevant microenvironment is a crucial aspect of preclinical animal models. Cancer comorbidities have been modeled in Oncopigs, including alcohol-induced fibrosis, a precursor to liver cancer development (29). Excessive alcohol intake is a major etiological factor for liver cancer, often leading to liver cirrhosis. The development of liver cancer models within a cirrhotic background holds immense importance for preclinical studies, given the profound effect of cirrhosis on tumor progression and treatment responses.

Alcohol-induced liver fibrosis was induced in Oncopigs via hepatic artery infusion of an emulsion of ethanol with ethiodized oil (29). Histological analysis revealed fibrosis and inflammation, with peak fibrosis severity at 8 weeks postinduction followed by liver recovery by week 20, likely due to the lack of persistent alcohol exposure (29). Transcriptional analysis of fibrotic liver tissue obtained at the peak of fibrosis demonstrated recapitulation of human cirrhosis hallmarks (30). Pathway alterations included cell proliferation, angiogenesis, extracellular matrix deposition, and oxidation-reduction processes. Furthermore, key alterations including activation of hepatic stellate cells, increased matrix metalloproteinase production, and altered expression of transporter genes involved in transport of anticancer drugs were observed, mimicking features seen in cirrhosis progression in humans (30). Liver cancer has been developed in Oncopigs harboring alcohol-induced fibrosis as described above, further enhancing the model's translational relevance (15, 19). Via concurrent modeling of cancer and cancer comorbidities, the value and translational relevance of Oncopigs for cancer studies can be enhanced significantly.

7. APPLICATIONS OF THE ONCOPIG CANCER MODEL

The development of new pharmacologic agents and devices for cancer treatment is a long, highly regulated, and expensive process. The average cancer drug takes more than a decade to develop, with costs exceeding $1 billion (31). The clinical trial process requires rigorous experimentation that effectively studies and documents instances of therapeutic failure and adverse events. The high costs and failure rates associated with clinical trials create an unmet clinical need for earlier therapeutic optimization and determination of therapeutic efficacy in a controlled and safe manner. As such, it is critical that researchers are able to investigate medications and interventional therapies on animal models prior to employing them in human clinical trials.

7.1. Applications of the Oncopig in Device Testing

Small animal models, including mice, rats, and rabbits, have significant limitations with respect to testing new devices and interventional therapies (32). To overcome the anatomical and size constraints inherent in these models, pigs have emerged as valuable alternatives for preclinical testing of devices and locoregional therapies. Testing locoregional therapies and devices in healthy pig liver models is a widely used and well-documented approach. For instance, Namur et al. (33) recently investigated doxorubicin concentrations after drug-eluting bead chemoembolization (DEB-TACE) in a porcine liver model, and Isfort et al. (34) used a porcine liver model to compare irreversible electroporation with and without the addition of DEB-TACE. Successes of in vivo porcine liver model investigations have spurred translational research in other pig organ models. Examples include assessing the safety and efficacy of microwave ablation in kidneys

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

(35, 36) and lungs (37), as well as investigating irreversible electroporation in the kidney (38) and histotripsy in the thyroid (39). These studies underscore the value of pig models for evaluating various types of catheter and percutaneous device–based therapies, as well as for exploring the efficacy of combination therapies.

Despite these advancements, a major limitation in these studies was the lack of a porcine tumor model to assess the respective applications. This limitation hampers the translation of research findings to clinical patients due to a lack of insights into therapeutic performance, clinical safety, and effectiveness. The Oncopig platform addresses this gap by enabling the targeting of clinically relevant tumors that can be created in a local environment that mimics human conditions, such as HCC in the setting of cirrhosis (29).

Since its development, the Oncopig has demonstrated substantial value in therapeutic investigations. For example, Nurili et al. (14) confirmed the value of Oncopigs for liver-directed locoregional therapy testing. In their study, liver tumors were induced in 18 Oncopigs, and treatment with catheter-directed transarterial embolization using 40–120-μm or 100–300-μm particles resulted in tumor size reduction, validating the utility of the Oncopig liver tumor model for therapeutic efficacy testing. Additional studies have supported the utility of the Oncopig liver cancer model for preclinical investigation of other contemporary therapies, such as transarterial radioembolization as a targeted internal brachytherapy for liver cancer (40). Other ongoing investigations using the Oncopig for thermal ablation, transarterial chemoembolization, and analytical and performance testing and preclinical validation of other clinically relevant technologies hold promise for the development of exciting new therapeutics for human clinical translation.

7.2. Applications of the Oncopig in Diagnostics

Radiologic imaging plays a crucial role in diagnosing cancer, monitoring tumor progression, and assessing treatment response. Leveraging porcine biomedical models, such as the Oncopig, can enhance the imaging processes by validating new imaging modalities and schemes for clinical translation. The ability to image large animal models with the same equipment used for humans offers a significant advantage for preclinical animal trials. With the rapid advancement of imaging modalities in clinical research, easily imaged large animal models are highly desirable for preclinical experimentation.

Diagnostic imaging investigations in pigs have yielded several advancements to date. For instance, Pirasteh et al. (41) validated a new imaging agent for liver fibrosis, ^{68}Ga-labeled fibroblast activation protein inhibitor (FAPI), by correlating liver ^{68}Ga-FAPI uptake with the histologic stage of liver fibrosis in five pig subjects. In a recent study, Gaba and colleagues (42) investigated the feasibility of magnetic resonance elastography (MRE) for staging Oncopig alcohol-induced liver fibrosis developed by transarterial administration of ethanol-ethiodized oil emulsion. MRE performed at one, two, and three months, with measurement of liver parenchymal stiffness, confirmed that MRE-quantifiable liver fibrosis could be induced in the Oncopig.

To further advance preclinical Oncopig imaging, future studies should focus on publishing imaging characteristics of tumors in large animal models. This includes defining normal from pathologic variants and developing standardized protocols for imaging acquisition and review. It is essential to validate tumors against human cancer imaging and develop standardized diagnostic and response assessment criteria tailored to large animal models. Universal adoption of large animal imaging protocols will validate their representational value and optimize their ability to recapitulate human disease (43).

Molecular marker detection is also promising for diagnosing, monitoring, and predicting treatment responses for many cancers (44–47). Blood-based screening is the preferred method

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

for monitoring biomarkers due to its feasibility, cost-effectiveness, and minimal patient risk. By allowing cancer detection at the point of care through the use of liquid biopsies, blood-based molecular marker testing has immense importance for improving cancer surveillance and effectively shifting the treatment timeline for disease detection. The Oncopig provides an excellent platform for developing new diagnostic technologies aimed at detecting circulating biomarkers. This approach is more feasible than using human subjects or tissues, as the Oncopig allows for knowledge of time zero of carcinogenesis and systematic phlebotomy and tissue acquisition for detection. Though additional studies are needed, this concept has been validated by Gaba et al. (15), who demonstrated that Oncopig HCC cell lines produced the clinical biomarker alpha fetoprotein at similar levels as human HCC cells, demonstrating the potential utility of serum alpha fetoprotein as a biomarker for tumor growth in the Oncopig HCC model, similar to use of this biomarker in clinical practice (48).

7.3. Applications of the Oncopig in Drug Testing

Although the size and anatomical resemblances between pigs and humans are crucial aspects of a porcine cancer model, equally significant are the genomic, epigenetic, physiological, metabolic, and immunological characteristics of pigs that mirror human biology (49, 50). These similarities confer clinical relevance to porcine models, enabling the testing of drug efficacy, drug delivery, and immunotherapy (51). The ability to induce tumors in Oncopigs with fully competent immune systems, along with the T cell–mediated antitumor immune response observed in Oncopig models of liver, pancreatic, and lung cancers, as described above, highlights the potential utility of Oncopigs for studying antitumor immune responses and preclinical testing of immunotherapies.

Gaba et al. (15) recently underscored the clinical relevance of the Oncopig for testing the efficacy of clinically relevant drugs. Their study demonstrated similar expression of key genes involved in drug metabolism and transport between Oncopig and human HCC. Furthermore, Oncopig HCC cells accurately predicted chemotherapeutic susceptibility of human HCC cells in vitro and outperformed murine HCC cells in predictive accuracy. Notably, six Oncopig HCC cell lines exhibited consistent susceptibility to the chemotherapy drugs sorafenib and doxorubicin, commonly used for systemic and locoregional HCC therapy, respectively (15). More recently, Segatto et al. (27) developed bladder cancer cell lines from Oncopigs and demonstrated consistent treatment responses between Oncopig and human cells to cisplatin, doxorubicin, and gemcitabine chemotherapeutics in vitro.

8. LIMITATIONS OF ONCOPIG CANCER MODELS

Despite the promise that Oncopigs hold for improving the clinical relevance of preclinical oncology studies and improving translation of novel drugs, devices, and diagnostic approaches into clinical practice, several limitations must be addressed for porcine biomedical models to reach their full potential. One major barrier for the use of Oncopigs and other porcine biomedical models is the high cost of porcine studies relative to those with mice, in addition to the need for dedicated large animal facilities. Importantly, although little can be done to address the increased cost compared to mouse studies, porcine studies are more cost effective than nonhuman primate and clinical studies. Moreover, even if immediately more costly, porcine studies may offer a superior value proposition compared with murine investigations if they are more prognostic of downstream outcomes in human patients, potentially allowing immense cost savings if negative clinical trials can be predicted and avoided. The need for dedicated large animal facilities represents another significant barrier for many researchers interested in performing porcine studies. Therefore,

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

national porcine biomedical research centers are needed to act as regional hubs, similar to the centers that were established to address the same need for the nonhuman primate research community to further drive model adoption.

In addition, porcine-specific resources, reagents, and tools are lacking compared to those available for humans and mice. For example, whereas researchers working with human and mouse samples can feasibly obtain antibodies validated to work in their respective species for Western blot, immunohistochemistry, or flow cytometry, very few available antibodies have been validated for use in pigs. Although members of the swine in biomedical research community have developed publicly available databases in which researchers can deposit information regarding antibodies validated for use in pigs (52), the availability of this database is not widely known. In addition, results supporting the cross-reactivity of these antibodies for pig proteins are not provided by the commercial entities supplying these antibodies. Therefore, many researchers are compelled to perform in-house validation work to confirm the information in the database before using it for studies. Specific to cancer modeling, the lack of porcinized immune checkpoint inhibitors also limits the utility of porcine cancer models for immunotherapeutic studies. Together, the significantly fewer commercially available plasmids, enzyme-linked immunosorbent assay kits, sequencing kits, RNAscope probes, microarrays, cytokines, chemokines, and growth factors available for pigs compared to humans and mice highlight the need to invest in porcine-specific reagents to address limitations associated with use of pigs in biomedical research.

Limitations specific to the Oncopig cancer model include the use of *KRAS* and *TP53* driver mutations. Although these mutations are observed in more than half of human cancers, they are not commonly observed in all cancer subtypes. A wide range of murine cancer models representative of a range of driver mutations and cancer types are readily available for studies, but increased costs and space requirements associated with maintaining transgenic pigs have resulted in development of broadly applicable porcine cancer models versus models tailored to specific cancer types or driver mutational profiles. Although these limitations can be overcome by using somatic cell editing approaches and cell type–specific Cre vectors, this represents an additional barrier that porcine models must overcome to facilitate further development of clinically relevant cancer subtypes.

Additional limitations of the Oncopig include the high inflammation rate within tumors, observed spontaneous tumor regression, and lack of observed metastatic disease. The presence of a strong inflammatory response concomitant with tumor development is a recurrent observation across various cancer types and induction methods (12, 14, 23, 25, 26). Although this inflammatory milieu represents an opportunity to better understand anticancer immunity, it presents challenges. Spontaneous tumor regression hinders long-term monitoring of tumor progression and therapeutic response. Additionally, the inability to distinguish between inflammatory and neoplastic lesions on imaging necessitates biopsy confirmation prior to treatment in preclinical studies. These limitations necessitate the incorporation of control groups with tumors but without treatment to discern spontaneous regression from treatment-related effects. As the use of the Oncopig model continues to grow, additional work will be required to understand the molecular mechanisms driving these phenotypes and develop approaches to address these limitations.

9. REFINEMENT OF ONCOPIG CANCER MODELS: FUTURE PROMISES

Since the development of the genetically engineered Oncopig Cancer Model nearly a decade ago, it has proven to be a valuable tool for modeling several types of cancer. Yet, ample room remains for refinement and expansion of Oncopig cancer models to enhance their applicability and translational relevance. Given the prevalence of *KRAS* and *TP53* mutations across multiple tumor

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

types, there is a promising opportunity to develop additional Oncopig cancer models beyond those currently established. Ongoing efforts to create models for glioblastoma and colorectal cancer in Oncopigs have shown promise, highlighting the importance of broadening the spectrum of cancer models to address unmet clinical needs.

One strategy to potentially enhance Oncopig cancer models is the use of in vivo viral vectors with tissue-specific Cre expression. A key limitation of many current Oncopig cancer models is the nonspecific transformation of various cell types resulting from the use of AdCre. By employing viral vectors that express Cre recombinase under the control of tissue-specific promoters, it is conceivable to refine cancer models by restricting transgene expression to target cells of interest. An exciting approach to further refine these models entails the use of viral vectors co-expressing Cre recombinase and CRISPR components. This dual functionality would enable simultaneous induction of transgene expression and specific gene mutations, thereby facilitating the development of genetically tailored tumors in Oncopigs. This innovative approach, previously employed in genetically modified mouse models (53), could provide a platform for rapid modeling of cooperative genetic events in cancer.

Another promising approach for efficient development of luminal tumors involves using a tumor-inducing patch capable of adhering to luminal surfaces and delivering Cre recombinase. These patches can administer Cre recombinase as well as combinations of oncogenic expression vectors and/or CRISPR components and can be applied to luminal tissue endoscopically. Ongoing research is investigating the use of tumor-inducing patches to induce tumors in Oncopigs by delivering Cre recombinase.

Another strategy that might improve Oncopig cancer models is the mitigation of lymphocytic inflammatory response frequently observed in Oncopig tumors. This can be achieved via the administration of glucocorticoids or immunosuppressives, although this may complicate the models and potentially impact their relevance.

Furthermore, there is potential for modeling cancer comorbidities through crossbreeding. For example, Ossabaw pigs—derived from a swine colony from Ossabaw Island near Georgia—are, through generations of natural selection, prone to gain large amounts of weight and develop obesity, insulin resistance, glucose intolerance (54), dyslipidemia (55, 56), and hypertension (55, 57), which are widely accepted characteristics of metabolic syndrome (57–59). Ossabaw swine fed a Western or nonalcoholic steatohepatitis (NASH) diet develop severe metabolic syndrome with a markedly abnormal liver histology that mimics human NASH, a quickly developing global epidemic (59). Crossbreeding Oncopigs with Ossabaw pigs could yield a metabolic model for NASH, diabetes, and obesity, offering a unique opportunity to study HCC in the setting of NASH liver disease microenvironment. Similar benefits could be derived from developing crosses between Oncopigs and pigs manifesting atherosclerotic disease (60) or other chronic health conditions.

10. CONCLUSION AND PERSPECTIVES

In conclusion, the Oncopig Cancer Model represents a significant advancement in preclinical cancer research, offering a versatile and clinically relevant platform for testing therapeutic interventions, particularly those not feasible to test in mouse models. By recapitulating key aspects of human cancer biology, including the expression of oncogenic transgenes *KRAS*^{G12D} and *TP53*^{R167H}, the Oncopig provides a platform for diverse cancer models.

Moving forward, there are several avenues for further exploration and enhancement of Oncopig cancer models. Collaborations between researchers from diverse disciplines, including oncology, genetics, pathology, interventional radiology, immunology, and veterinary medicine,

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

will be instrumental in maximizing the potential of the Oncopig model and accelerating progress in cancer research. By leveraging the strengths of multidisciplinary approaches, researchers can address complex challenges in cancer biology and develop innovative strategies for cancer prevention, diagnosis, and treatment. Continued efforts to expand the repertoire of cancer models in Oncopigs, including the development of models for additional cancer types, and the refinement of existing models, will contribute to increased applications and translational relevance of Oncopig studies of therapeutics. Additionally, the use of Oncopig models for broader applications, such as testing emerging therapies and refining imaging techniques, holds promise for advancing cancer research and clinical translation.

Overall, the Oncopig Cancer Model represents an invaluable tool for bridging the gap between preclinical research and clinical trials, offering insights that can ultimately improve outcomes for cancer patients. With ongoing innovation and collaboration, the Oncopig model could drive transformative advancements in cancer research and therapeutic development in the years to come.

FUTURE POTENTIAL TO REFINE AND EXPAND ONCOPIG CANCER MODELS

1. Additional cancer models beyond established ones are needed to address unmet clinical needs.
2. Transgene expression could be induced using viral vectors with tissue-specific Cre expression.
3. Viral vectors that coexpress Cre recombinase and CRISPR components could be used to induce transgene expression and additional defined gene mutations.
4. Tumor-inducing patches that administer Cre recombinase could be used for induction of luminal tumors.
5. Tumor regression can be prevented by mitigation of the inflammatory antitumor immune response.
6. Oncopigs can be crossbred with pigs prone to developing comorbidities, such as pigs with metabolic syndrome or atherosclerosis.

DISCLOSURE STATEMENT

L.E. reports grants from the US National Institutes of Health (NIH) and the Society of Interventional Radiology and a pending patent for Patch for Targeted Delivery of an Oncogenic Cargo to a Tissue, US Patent Appl. No. 63/306,449. R.C.G. reports grants from the NIH, Guerbet USA LLC, Janssen Research & Development LLC, NeoTherma Oncology, TriSalus Life Sciences, and Sus Clinicals, Inc.; consulting fees from Boston Consulting Group; a pending patent for Modeling Oncology on Demand, US Provisional Patent Appl. No. 62/813,307; participation on Sus Clinicals, Inc. Scientific Advisory Board; participation on Fluidx Medical Technology Advisory Board; participation on Kaveri University Advisory Board; and stock or stock options in Sus Clinicals, Inc. M.M.N. reports grants from the NIH, the Society of Interventional Radiology, Janssen Research & Development LLC, Earli Inc., and Sus Clinicals, Inc., and stock or stock options in Sus Clinicals, Inc. L.B.S. and K.M.S. report grants from the NIH, US Department of Agriculture, Society of Interventional Radiology, Guerbet USA LLC, Janssen Research & Development LLC, TriSalus Life Sciences, and Earli Inc. L.B.S. and K.M.S. work for Sus Clinicals, which provides the Oncopig and other pig-based preclinical testing services to customers; they have pending patents for Modeling Oncology on Demand, US Provisional Patent Appl. No. 62/813,307, and Patch for Targeted Delivery of an Oncogenic Cargo to a Tissue, US Patent Appl. No. 63/306,449.

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

LITERATURE CITED

1. Sung H, Ferlay J, Siegel RL, Laversanne M, Soerjomataram I, et al. 2021. Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA Cancer J. Clin.* 71:209–49
2. Honkala A, Malhotra SV, Kummar S, Junttila MR. 2022. Harnessing the predictive power of preclinical models for oncology drug development. *Nat. Rev. Drug Discov.* 21:99–114
3. Bryda EC. 2013. The mighty mouse: the impact of rodents on advances in biomedical research. *Mo. Med.* 110:207–11
4. Onaciu A, Munteanu R, Munteanu VC, Gulei D, Raduly L, et al. 2020. Spontaneous and induced animal models for cancer research. *Diagnostics* 10:660
5. Sajjad H, Imtiaz S, Noor T, Siddiqui YH, Sajjad A, Zia M. 2021. Cancer models in preclinical research: a chronicle review of advancement in effective cancer research. *Anim. Model Exp. Med.* 4:87–103
6. Schook L, Beattie C, Beever J, Donovan S, Jamison R, et al. 2005. Swine in biomedical research: creating the building blocks of animal models. *Anim. Biotechnol.* 16:183–90
7. Hou N, Du X, Wu S. 2022. Advances in pig models of human diseases. *Anim. Model Exp. Med.* 5:141–52
8. Meyerholz DK, Burrough ER, Kirchhof N, Anderson DJ, Helke KL. 2024. Swine models in translational research and medicine. *Vet. Pathol.* 61:512–23
9. Schook LB, Collares TV, Darfour-Oduro KA, De AK, Rund LA, et al. 2015. Unraveling the swine genome: implications for human health. *Annu. Rev. Anim. Biosci.* 3:219–44
10. Groenen MA, Archibald AL, Uenishi H, Tuggle CK, Takeuchi Y, et al. 2012. Analyses of pig genomes provide insight into porcine demography and evolution. *Nature* 491:393–98
11. Schook LB, Collares TV, Hu W, Liang Y, Rodrigues FM, et al. 2015. A genetic porcine model of cancer. *PLOS ONE* 10:e0128864
12. Overgaard NH, Principe DR, Schachtschneider KM, Jakobsen JT, Rund LA, et al. 2018. Genetically induced tumors in the Oncopig model invoke an antitumor immune response dominated by cytotoxic $CD8\beta^+$ T cells and differentiated $\gamma\delta$ T cells alongside a regulatory response mediated by $FOXP3^+$ T cells and immunoregulatory molecules. *Front. Immunol.* 9:1301
13. Llovet JM, De Baere T, Kulik L, Haber PK, Greten TF, et al. 2021. Locoregional therapies in the era of molecular and immune treatments for hepatocellular carcinoma. *Nat. Rev. Gastroenterol. Hepatol.* 18:293–313
14. Nurili F, Monette S, Michel AO, Bendet A, Basturk O, et al. 2021. Transarterial embolization of liver cancer in a transgenic pig model. *J. Vasc. Interv. Radiol.* 32:510–17.e3
15. Gaba RC, Elkhadragy L, Boas FE, Chaki S, Chen HH, et al. 2020. Development and comprehensive characterization of porcine hepatocellular carcinoma for translational liver cancer investigation. *Oncotarget* 11:2686–701
16. Schachtschneider KM, Schwind RM, Darfour-Oduro KA, De AK, Rund LA, et al. 2017. A validated, transitional and translational porcine model of hepatocellular carcinoma. *Oncotarget* 8:63620–34
17. Patel SS, Sandur A, El-Kebir M, Gaba RC, Schook LB, Schachtschneider KM. 2021. Transcriptional profiling of porcine HCC xenografts provides insights into tumor cell microenvironment signaling. *Front. Genet.* 12:657330
18. Elkhadragy L, Dasteh Goli K, Totura WM, Carlino MJ, Regan MR, et al. 2022. Effect of CRISPR knockout of AXIN1 or ARID1A on proliferation and migration of porcine hepatocellular carcinoma. *Front. Oncol.* 12:904031
19. Elkhadragy L, Carlino MJ, Jordan LR, Schook LB, Gaba RC, Schachtschneider KM. 2021. Development of a precision hepatocellular carcinoma model in transgenic pigs by *ex vivo* CRISPR editing. *Hepatology* 74(Suppl.):684A
20. Cancer Genome Atlas Res. Netw. 2017. Integrated genomic characterization of pancreatic ductal adenocarcinoma. *Cancer Cell* 32:185–203.e13
21. Connor AA, Denroche RE, Jang GH, Lemire M, Zhang A, et al. 2019. Integration of genomic and transcriptional features in pancreatic cancer reveals increased cell cycle progression in metastases. *Cancer Cell* 35:267–82.e7

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

22. Principe DR, Overgaard NH, Park AJ, Diaz AM, Torres C, et al. 2018. KRAS^{G12D} and TP53^{R167H} cooperate to induce pancreatic ductal adenocarcinoma in *Sus scrofa* pigs. *Sci. Rep.* 8:12548
23. Mondal P, Patel NS, Bailey K, Aravind S, Cartwright SB, et al. 2023. Induction of pancreatic neoplasia in the *KRAS/TP53* Oncopig. *Dis. Model Mech.* 16:dmm049699
24. Boas FE, Nurili F, Bendet A, Cheleuitte-Nieves C, Basturk O, et al. 2020. Induction and characterization of pancreatic cancer in a transgenic pig model. *PLOS ONE* 15:e0239391
25. Ghosn M, Elsakka AS, Petre EN, Cheleuitte-Nieves C, Tammela T, et al. 2023. Induction and preliminary characterization of neoplastic pulmonary nodules in a transgenic pig model. *Lung Cancer* 178:157–65
26. Aulitzky A, Wilcox Vanden Berg R, Kim K, Al-Ahmadie H, Monette S, Coleman JA. 2023. Development of a porcine model of bladder cancer using the Oncopig. *J. Urol.* 209(Suppl. 4):e182
27. Segatto NV, Simoes LD, Bender CB, Sousa FS, Oliveira TL, et al. 2024. Oncopig bladder cancer cells recapitulate human bladder cancer treatment responses *in vitro*. *Front. Oncol.* 14:1323422
28. Elkhadragy L, Regan MR, Totura WM, Goli KD, Patel S, et al. 2020. Generation of genetically tailored porcine liver cancer cells by CRISPR/Cas9 editing. *BioTechniques* 70:37–48
29. Gaba RC, Mendoza-Elias N, Regan DP, Garcia KD, Lokken RP, et al. 2018. Characterization of an inducible alcoholic liver fibrosis model for hepatocellular carcinoma investigation in a transgenic porcine tumorigenic platform. *J. Vasc. Interv. Radiol.* 29:1194–202.e1
30. Yasmin A, Regan DP, Schook LB, Gaba RC, Schachtschneider KM. 2021. Transcriptional regulation of alcohol induced liver fibrosis in a translational porcine hepatocellular carcinoma model. *Biochimie* 182:73–84
31. DiMasi JA, Grabowski HG, Hansen RW. 2016. Innovation in the pharmaceutical industry: new estimates of R&D costs. *J. Health Econ.* 47:20–33
32. Aravalli RN, Golzarian J, Cressman EN. 2009. Animal models of cancer in interventional radiology. *Eur. Radiol.* 19:1049–53
33. Namur J, Wassef M, Millot JM, Lewis AL, Manfait M, Laurent A. 2010. Drug-eluting beads for liver embolization: concentration of doxorubicin in tissue and in beads in a pig model. *J. Vasc. Interv. Radiol.* 21:259–67
34. Isfort P, Rauen P, Na HS, Ito N, von Stillfried S, et al. 2019. Does drug-eluting bead TACE enhance the local effect of IRE? Imaging and histopathological evaluation in a porcine model. *Cardiovasc. Intervent. Radiol.* 42:880–85
35. Sommer CM, Arnegger F, Koch V, Pap B, Holzschuh M, et al. 2012. Microwave ablation of porcine kidneys in vivo: effect of two different ablation modes ("temperature control" and "power control") on procedural outcome. *Cardiovasc. Intervent. Radiol.* 35:653–60
36. Laeseke PF, Lee FT Jr., Sampson LA, van der Weide DW, Brace CL. 2009. Microwave ablation versus radiofrequency ablation in the kidney: High-power triaxial antennas create larger ablation zones than similarly sized internally cooled electrodes. *J. Vasc. Interv. Radiol.* 20:1224–29
37. Carberry GA, Nocerino E, Cristescu MM, Smolock AR, Lee FT Jr., Brace CL. 2017. Microwave ablation of the lung in a porcine model: vessel diameter predicts pulmonary artery occlusion. *Cardiovasc. Intervent. Radiol.* 40:1609–16
38. Wagstaff PG, de Bruin DM, van den Bos W, Ingels A, van Gemert MJ, et al. 2015. Irreversible electroporation of the porcine kidney: temperature development and distribution. *Urol. Oncol.* 33:168.e1–7
39. Swietlik JF, Mauch SC, Knott EA, Zlevor A, Longo KC, et al. 2021. Noninvasive thyroid histotripsy treatment: proof of concept study in a porcine model. *Int. J. Hyperth.* 38:798–804
40. Schachtschneider KM, Arepally A, Blanco D, Hussey D, Mehta V, et al. 2024. Preclinical evaluation of Yttrium-90 radioembolization in the Oncopig liver cancer model. *J. Vasc. Interv. Radiol.* 35(3 Suppl.):S226
41. Pirasteh A, Periyasamy S, Meudt JJ, Liu Y, Lee LM, et al. 2022. Staging liver fibrosis by fibroblast activation protein inhibitor PET in a human-sized swine model. *J. Nucl. Med.* 63:1956–61
42. Gaba RC, Elkhadragy L, Pennix T, Schachtschneider KM, Bolt CR, et al. 2024. Magnetic resonance elastography for staging liver fibrosis in the Oncopig. *Diagnostics* 14(17):1880
43. Faraji F, Gaba RC. 2019. Radiologic modalities and response assessment schemes for clinical and preclinical oncology imaging. *Front. Oncol.* 9:471

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25

44. Behne T, Copur MS. 2012. Biomarkers for hepatocellular carcinoma. *Int. J. Hepatol.* 2012:859076
45. van Veldhoven K, Polidoro S, Baglietto L, Severi G, Sacerdote C, et al. 2015. Epigenome-wide association study reveals decreased average methylation levels years before breast cancer diagnosis. *Clin. Epigenetics* 7:67
46. Lin KT, Shann YJ, Chau GY, Hsu CN, Huang CY. 2014. Identification of latent biomarkers in hepatocellular carcinoma by ultra-deep whole-transcriptome sequencing. *Oncogene* 33:4786–94
47. Ganepola GA, Nizin J, Rutledge JR, Chang DH. 2014. Use of blood-based biomarkers for early diagnosis and surveillance of colorectal cancer. *World J. Gastrointest. Oncol.* 6:83–97
48. Lou J, Zhang L, Lv S, Zhang C, Jiang S. 2017. Biomarkers for hepatocellular carcinoma. *Biomark. Cancer* 9:1370
49. Schachtschneider KM, Schwind RM, Newson J, Kinachtchouk N, Rizko M, et al. 2017. The Oncopig Cancer Model: an innovative large animal translational oncology platform. *Front. Oncol.* 7:190
50. Overgaard NH, Fan TM, Schachtschneider KM, Principe DR, Schook LB, Jungersen G. 2018. Of mice, dogs, pigs, and men: choosing the appropriate model for immuno-oncology research. *ILAR J.* 59:247–62
51. Bailey KL, Carlson MA. 2019. Porcine models of pancreatic cancer. *Front. Oncol.* 9:144
52. Dawson HD, Lunney JK. 2018. Porcine cluster of differentiation (CD) markers 2018 update. *Res. Vet. Sci.* 118:199–246
53. Sanchez-Rivera FJ, Papagiannakopoulos T, Romero R, Tammela T, Bauer MR, et al. 2014. Rapid modelling of cooperating genetic events in cancer through somatic genome editing. *Nature* 516:428–31
54. Wangsness PJ, Martin RJ, Gahagan JH. 1977. Insulin and growth hormone in lean and obese pigs. *Am. J. Physiol.* 233: E104–8
55. Edwards JM, Neeb ZP, Alloosh MA, Long X, Bratz IN, et al. 2010. Exercise training decreases store-operated Ca^{2+} entry associated with metabolic syndrome and coronary atherosclerosis. *Cardiovasc. Res.* 85:631–40
56. Etherton TD, Kris-Etherton PM. 1980. Characterization of plasma lipoproteins in swine with different propensities for obesity. *Lipids* 15:823–29
57. Neeb ZP, Edwards JM, Alloosh M, Long X, Mokelke EA, Sturek M. 2010. Metabolic syndrome and coronary artery disease in Ossabaw compared with Yucatan swine. *Comp. Med.* 60:300–15
58. Dyson MC, Alloosh M, Vuchetich JP, Mokelke EA, Sturek M. 2006. Components of metabolic syndrome and coronary artery disease in female Ossabaw swine fed excess atherogenic diet. *Comp. Med.* 56:35–45
59. Lee L, Alloosh M, Saxena R, Van Alstine W, Watkins BA, et al. 2009. Nutritional model of steatohepatitis and metabolic syndrome in the Ossabaw miniature swine. *Hepatology* 50:56–67
60. Namous H, Strillacci MG, Braz CU, Shanmuganayagam D, Krueger C, et al. 2023. ITGB2 is a central hub-gene associated with inflammation and early fibro-atheroma development in a swine model of atherosclerosis. *Atheroscler. Plus* 54:30–41

Downloaded from www.annualreviews.org. Guest (guest) IP: 211.177.141.75 On: Sun, 02 Nov 2025 11:15:25